

Please if you haven't filled out the when2meet do it before looking at the document:

<https://www.when2meet.com/?28900063-pKkhR>

Next Week/Important Information:

- March 5, 2025 is the presentations for the beginner projects
- 2 maybe 3 more meetings left
 - In week 4, we will finish the remaining code and start with the slides distribution to present the information and split up the work (**please be present for this next meeting if you want a say in which portion of the presentation you want**)
 - In week 5, we will practice presenting the slides (by week 5 before our fifth meeting, have your portion done of the slides be done)
 - In week 6, depending on how meeting 5 will go and if we finish everything, there may or may not be a meeting that week
 - The presentation will be approx 5 minutes long in total

Week 3 Notes Recap:

We are now fine tuning the data set : feature selection

Let's look at adjusted R-squared Values

(First Code Chunk) full model:

```
```{r}
Build a full model and perform a F-test on it
insurance_full_model <- lm(premium ~ ., data = insurance_model_data)
summary(insurance_full_model)
```
```

Call:

```
lm(formula = premium ~ ., data = insurance_model_data)
```

Residuals:

| Min | 1Q | Median | 3Q | Max |
|----------|---------|--------|--------|---------|
| -11338.4 | -2839.7 | -923.3 | 1403.0 | 29746.6 |

Coefficients:

| | Estimate | Std. Error | t value | Pr(> t) |
|-------------|-----------|------------|---------|------------|
| (Intercept) | -12924.45 | 1169.26 | -11.053 | <2e-16 *** |
| age | 261.01 | 13.10 | 19.926 | <2e-16 *** |
| bmi | 327.97 | 30.99 | 10.584 | <2e-16 *** |
| children | 424.09 | 154.64 | 2.742 | 0.0062 ** |
| smoker | 23602.62 | 449.57 | 52.501 | <2e-16 *** |
| region | 232.74 | 167.43 | 1.390 | 0.1648 |

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5990 on 1064 degrees of freedom

Multiple R-squared: 0.7585, Adjusted R-squared: 0.7574

F-statistic: 668.5 on 5 and 1064 DF, p-value: < 2.2e-16

we now build a full model and do an F-test to see if the independent variables are truly significant or not. Based on our information we will try to build a smaller but just as effective model. The R-statistic and adjusted R-statistic of 0.75 are both pretty high however our children and region variables show less significance when compared to other independent variables.

We are interested in the Coefficients Table (this is the really important part):

Let's break this down:

"Estimate" is how much each factor affects the premium:

- Age: Each year older adds \$261 to your premium
- BMI: Each point higher adds \$328
- Children: Each child adds \$424
- Smoking: Being a smoker adds a whopping \$23,603!
- Region: Changes premium by \$233 (but we're not sure about this one)

Those stars (***) are like confidence ratings:

- More stars = more confident we are about that factor
- No stars (like with region) means we're unsure

What can we take from the code:

- Smoking is BY FAR the biggest factor (adds \$23,603)
- Age, BMI, and number of children all matter too
- Region doesn't seem to matter much (no stars, high p-value)

Regular R-squared vs. Adjusted R-squared:

- Regular R-squared (R^2):
- Measures how well your model fits the data
- Ranges from 0 to 1 (or 0% to 100%)
- Example: R^2 of 0.7585 means our model explains 75.85% of the variation in insurance premiums
- BUT there's a catch: R^2 will always increase when you add more variables, even if they're not helpful!

Adjusted R-squared:

- It's a "corrected" version of R^2
- Penalizes you for adding unnecessary variables
- Can actually decrease if you add unhelpful variables
- More realistic measure of model performance

(Second Code Chunk) Removing region code:

```

## {r}
#Reduced Model 1: Exclude region
reduced_model_no_region <- lm(premium ~ age + bmi + smoker + children, data = insurance_model_data)
summary(reduced_model_no_region)

Call:
lm(formula = premium ~ age + bmi + smoker + children, data = insurance_model_data)

Residuals:
    Min       1Q   Median       3Q      Max
-11766.4  -2935.6   -903.2   1387.2  29611.8

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -12179.79    1039.78  -11.714  <2e-16 ***
age           261.48       13.10    19.959  <2e-16 ***
bmi           322.21       30.72    10.488  <2e-16 ***
smoker       23612.13     449.71    52.505  <2e-16 ***
children      416.32      154.60     2.693   0.0072 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5992 on 1065 degrees of freedom
Multiple R-squared:  0.7581,    Adjusted R-squared:  0.7572
F-statistic: 834.4 on 4 and 1065 DF,  p-value: < 2.2e-16

```

When region is excluded, the Adjusted R-squared value slightly decreases from 0.7574 to 0.7572, indicating a negligible impact on the model's explanatory power. The residual standard error increases marginally to 5992.

Removing region from the full model since it suggests to have little impact on the dataset output, we now are comparing two models adjusted R-squared Values: Full Model (with region) and Removed Model (without region)

- Full model (with region): Adjusted $R^2 = 0.7574$
- Without region: Adjusted $R^2 = 0.7572$

The tiny drop (0.0002) tells us the region wasn't adding much value.

This is why Adjusted R^2 is better for comparing models: it helps us find the sweet spot between:

- Having enough variables to make good predictions
- Not having so many variables that we're just adding noise

What This All Means:

1. Removing region barely affected our predictions
2. All our remaining variables are still significant (they have stars)
3. The relationships (like smoking adding \$23,612) stayed almost exactly the same
4. Our prediction errors (residuals) barely changed

(Third Code Chunk) Removing Children & Region:

```

##{r}
#Reduced Model 2: Exclude both region and children
reduced_model_no_region_children <-lm(premium~ age + bmi + smoker, data = insurance_model_data)
summary(reduced_model_no_region_children)

```

```

Call:
lm(formula = premium ~ age + bmi + smoker, data = insurance_model_data)

```

```

Residuals:
    Min       1Q   Median       3Q      Max
-12211.2  -2979.3   -962.1   1439.3  29127.3

```

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -11877.12    1036.71  -11.46  <2e-16 ***
age           263.01      13.13    20.04  <2e-16 ***
bmi           324.82      30.80    10.55  <2e-16 ***
smoker       23645.45     450.86    52.45  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 6010 on 1066 degrees of freedom
Multiple R-squared:  0.7565,    Adjusted R-squared:  0.7558
F-statistic: 1104 on 3 and 1066 DF,  p-value: < 2.2e-16

```

Excluding both region and children results in a more noticeable drop in Adjusted R-squared, from 0.7574 in the full model to 0.7558. The residual standard error increases slightly to 6010.

Key Changes:

- Original Adjusted R-squared: 0.7574
- After removing region: 0.7572
- After removing both: 0.7558
- This is a bigger drop (0.0016)

What This Tells Us:

- The drop in predictive power is more noticeable
- All remaining variables are still highly significant (***)
- Prediction errors increased more substantially
- The model is simpler but at a greater cost to accuracy

Unlike removing regions, removing children does hurt our model's performance enough to matter. While the model is simpler, we're losing meaningful predictive power. It's like removing an important ingredient from the recipe - the dish doesn't taste quite as good anymore.

Therefore, based on this analysis, we should:

- Keep age, BMI, smoker, and children
- Remove only region
- This gives us the best balance of simplicity and accuracy

F TESTS Analyzation:

Why are we now looking at F-value and not stopping at just the adjusted R-value ?

Adjusted R-squared:

- Tells us about overall model performance
- Shows us how much variation we can explain
- Example: dropping from 0.7574 to 0.7572 (for region)

F-test:

- Tells us if the change is statistically significant
- Accounts for both the size of the change AND sample size
- Helps us know if changes happened by chance
- Example: $F=1.9322$, $p=0.1648$ confirms region removal is okay

How to interpret F-Tests:

-  Remove variable ($p > 0.05$)
-  Don't remove variable ($p < 0.05$)

How to interpret F-Values:

- F-value close to 1: The variable(s) probably aren't important
- F-value much larger than 1: The variable(s) are probably important
- The larger the F-value, the stronger the evidence that the variables matter

(Fourth Code Chunk) First F-Test :

Let me break down these F-tests and explain what they're telling us:

```
##{r}
#F-test: Full model vs. Reduced Model 1 (No Region)
anova(reduced_model_no_region, insurance_full_model)
```

Analysis of Variance Table

```
Model 1: premium ~ age + bmi + smoker + children
Model 2: premium ~ age + bmi + children + smoker + region
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1   1065 3.8240e+10
2   1064 3.8171e+10  1  69318429 1.9322 0.1648
```

The F-test comparing this reduced model to the full model results in $F = 1.9322$ with $p = 0.1648$, showing no statistically significant difference between the two models. This confirms that region is not a meaningful predictor and can be excluded without affecting the model's performance.

Key Results:

- F-value = 1.9322
- p-value = 0.1648

What Does this mean :

- We're comparing two models: one with region and one without
- The p-value (0.1648) is much larger than 0.05
- Think of 0.05 as our "threshold for caring"
- This is like saying "there's only a 16.48% chance that region matters"
- Therefore, we can safely remove the region!

F-value = 1.9322

What this means:

- A low F-value like 1.9322 suggests removing region didn't make the model much worse
- The closer to 1, the less impact removing the variable had
- Think of it as: "removing region only made our predictions about 1.9 times worse than the natural variation in our data"

(Fifth Code Chunk) Second F-Test:

```
## {r}
#F-test: Full model vs. Reduced Model 2 (No Region and Children)
anova(reduced_model_no_region_children, insurance_full_model)
```

Analysis of Variance Table

```
Model 1: premium ~ age + bmi + smoker
Model 2: premium ~ age + bmi + children + smoker + region
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1   1066 3.8501e+10
2   1064 3.8171e+10  2 329679500 4.5948 0.01031 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The F-test comparing this reduced model to the full model results in $F = 1.9322$ with $p = 0.1648$, showing no statistically significant difference between the two models. This confirms that region is not a meaningful predictor and can be excluded without affecting the model's performance.

Key Results:

- F-value = 4.5948
- p-value = 0.01031

What This Means:

- Now we're comparing the simplest model (no region, no children) with the full model
- The p-value (0.01031) is less than 0.05
- This is like saying "there's a 98.97% chance that these variables matter"
- The * means "this difference is significant"
- Therefore, we should NOT remove children from the model

F-value = 4.5948

What this means:

- A higher F-value of 4.5948 indicates a bigger impact
- Removing both variables made the model noticeably worse
- Think of it as: "removing both variables made our predictions about 4.6 times worse than the natural variation in our data"

Based on our systematic analysis, here's our final model recommendation:

Final Model: `premium ~ age + bmi + children + smoker`

Here's why this is our best model:

Variables We Kept:

- Age (adds \$261 per year)
- BMI (adds \$322 per BMI point)
- Children (adds \$416 per child)
- Smoker (adds \$23,612 if yes)

Variable We Removed:

- Region (removed because):
- P-value was 0.1648 (> 0.05)
- F-test showed no significant difference ($F = 1.9322$)
- Adjusted R-squared barely changed ($0.7574 \rightarrow 0.7572$)